EDS Lab Assignments

Practical 2

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3.1.1 Numpy Array Operations

Problem Statement:

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Code:

```
import numpy as np
rows, cols = map(int, input().split()) elements
= []

for _ in range (rows):
        elements.extend(map(int, input().split()))

array = np.array(elements).reshape(rows, cols)
print(array) print(array.ndim) print(array.shape)
print(array.size)
```

Output:

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3.1.1. Numpy Array Operations

10:29

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases +

Explorer

Average time	Maximum time
0.015 s	0.048 s
14.83 ms	48.00 ms

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 20 ms Debug

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]	[5 6 7 8]
[9 10 11 12]]	[9 10 11 12]]
2	2
(3, 4)	(3, 4)
12	12

Test case 2 4 ms Debug

Expected output	Actual output
0 0	0 0
[]	[]
2	2
(0, 0)	(0, 0)
0	0

Debugger

Terminal Test cases

‹ Prev Reset Submit Next ›

3.2.1 Numpy: Matrix Operations

Problem Statement:

The given code takes two 3×3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (matrix_a + matrix_b)
2. Subtraction (matrix_a – matrix_b)
3. Element-wise Multiplication (matrix_a * matrix_b)
4. Matrix Multiplication (matrix_a · matrix_b)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

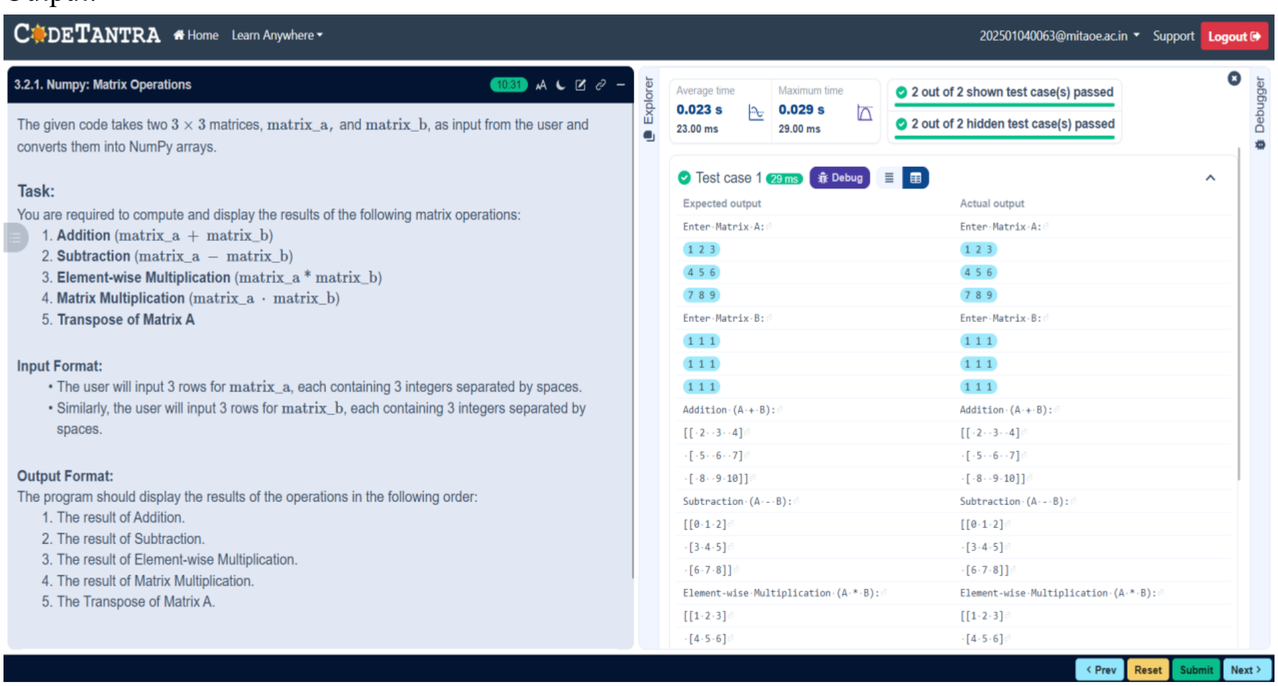
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np

# Input matrices print("Enter
Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
addition_result = matrix_a + matrix_b print("Addition
(A + B):")
```

```
print(addition_result)

# Subtraction
subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(subtraction_result)

# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):") print(elementwise_multiplication_result)

# Matrix multiplication (dot product)
matrix_multiplication_result = np.dot(matrix_a, matrix_b) print("A
dot B:")
print(matrix_multiplication_result)

# Transpose transpose_a
= matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

Output:



3.2.2 Numpy: Horizontal and Vertical Stacking of Arrays

Problem Statement:

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

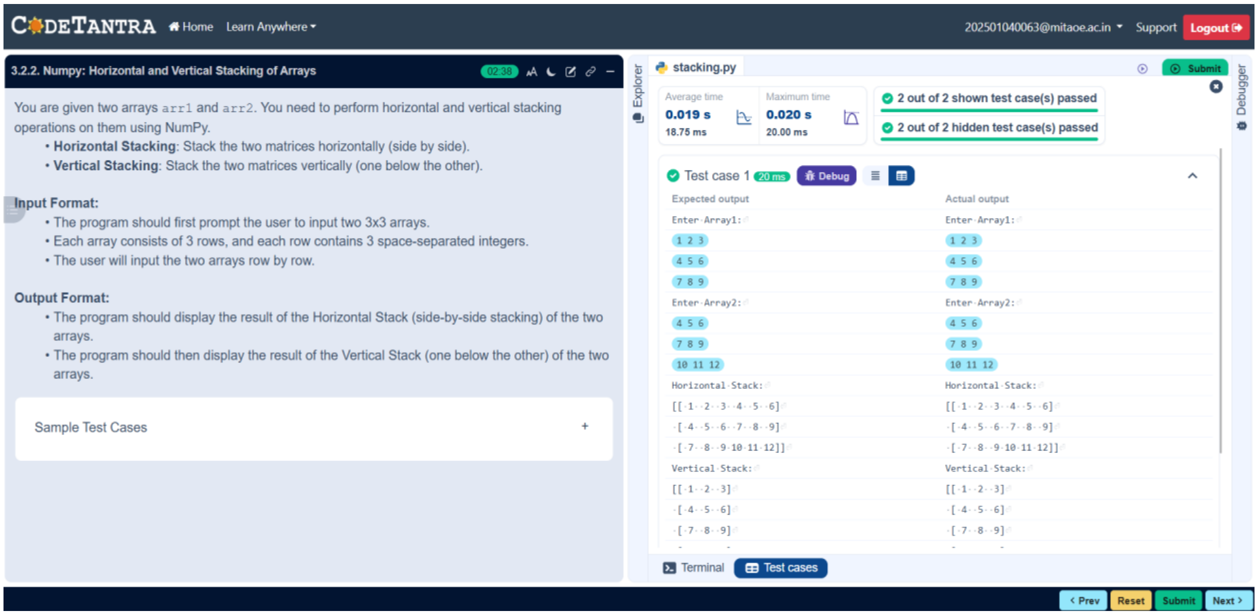
Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices print("Enter
Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
a = np.hstack((arr1, arr2))
print("Horizontal  Stack:")
print(a)

# Perform vertical stacking (vstack)
b = np.vstack((arr1, arr2))
print("Vertical Stack:") print(b)
```

Output:



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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays
You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.
• Horizontal Stacking: Stack the two matrices horizontally (side by side).
• Vertical Stacking: Stack the two matrices vertically (one below the other).
Input Format:
• The program should first prompt the user to input two 3x3 arrays.
• Each array consists of 3 rows, and each row contains 3 space-separated integers.
• The user will input the two arrays row by row.
Output Format:
• The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
• The program should then display the result of the Vertical Stack (one below the other) of the two arrays.
Sample Test Cases
stacking.py
Average time 0.019 s 18.75 ms
Maximum time 0.020 s 20.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1 20 ms
Debug
Expected output
Actual output
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[1 2 3 4 5 6]
[4 5 6 7 8 9]
[7 8 9 10 11 12]]
Vertical Stack:
[[1 2 3]
[4 5 6]
[7 8 9]
Terminal
Test cases
Prev
Reset
Submit
Next

3.2.3 Numpy: Custom Sequence Generation

Problem Statement:

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

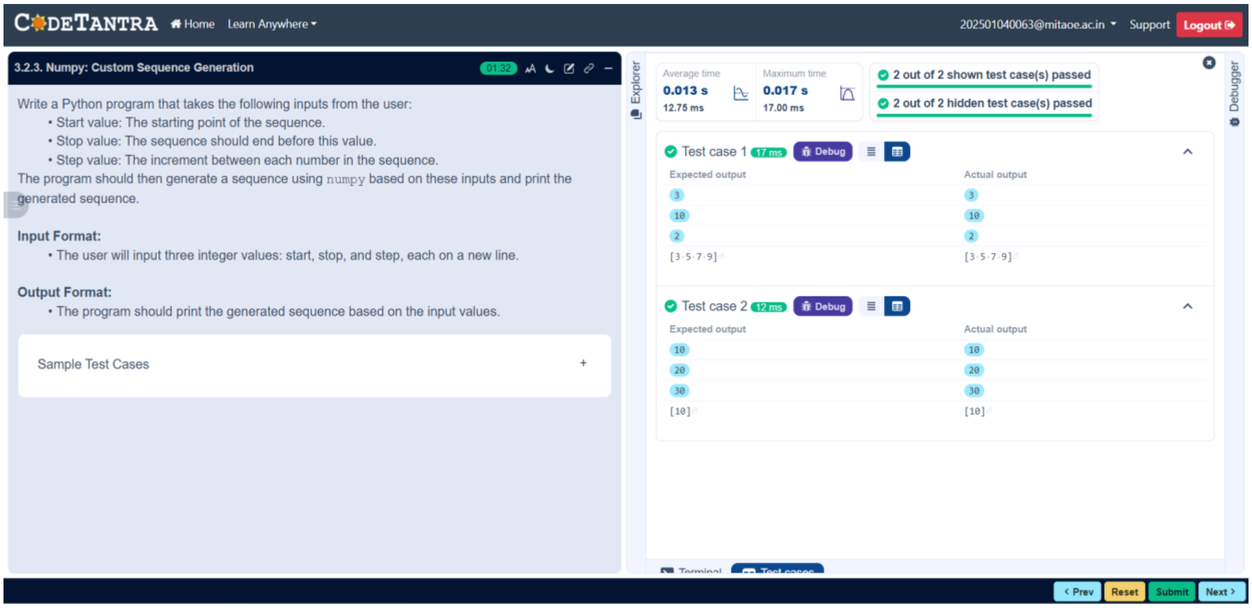
Output Format:

- The program should print the generated sequence based on the input values.

Code:

```
import numpy as np
# Take user input for the start, stop, and step of the sequence
start = int(input()) stop = int(input()) step = int(input())
# Generate the sequence using np.arange()
a = np.arange(start, stop, step) #
Print the generated sequence
print(a)
```

Output:



3.2.4 Numpy: Arithmetic and Statistical Operations, Mathematical Operations and Bitwise Operators

Problem Statement:

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:
   - Compute the element-wise sum, di erence, and product of the two arrays.
2. Statistical Operations:
   - Calculate the mean, median, and standard deviation of array A.
3. Bitwise Operations:
   - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

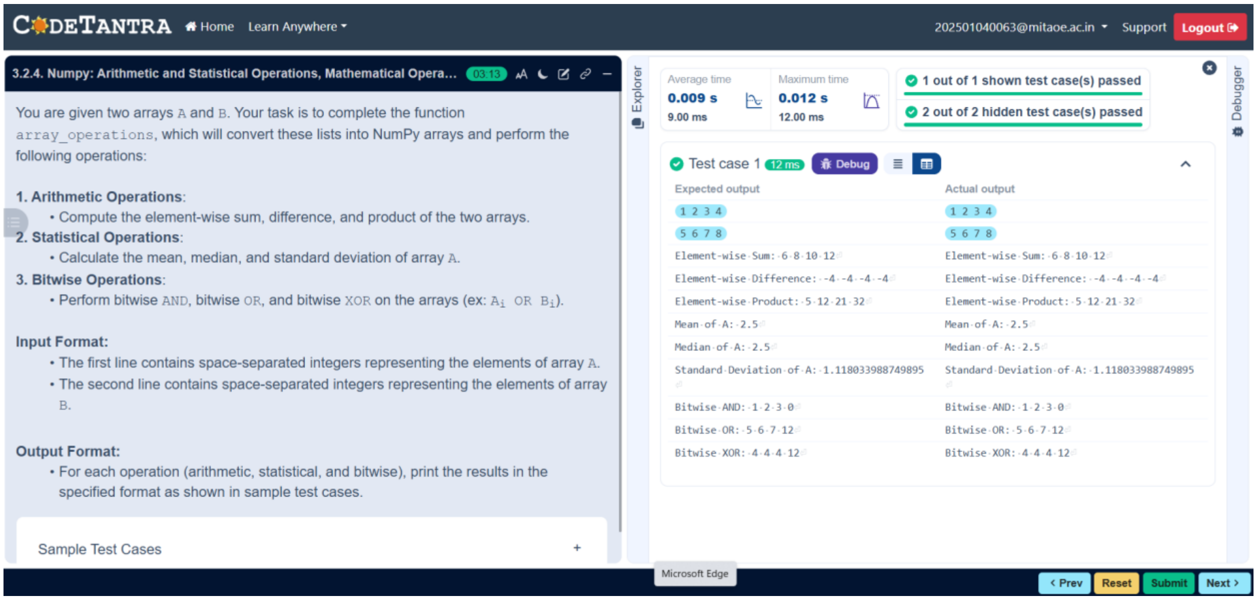
Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np

def array_operations(A, B):
            # Convert A and B to NumPy arrays
              A= np.array(A)
              B= np.array(B)

          # Arithmetic Operations
sum_result = A + B      di _result
= A - B
            prod_result = A * B

          # Statistical Operations
mean_A = np.mean(A)           median_A
= np.median(A)          std_dev_A =
np.std(A)

          # Bitwise Operations
and_result = A & B      or_result = A |
B
            xor_result = A ^ B

   # Output results with one space between each element         print("Element-
wise Sum:", ' '.join(map(str, sum_result)))       print("Element-wise Dierence:", '
'.join(map(str, di           _result)))         print("Element-wise Product:", '
'.join(map(str, prod_result)))

          print(f"Mean of A: {mean_A}")
print(f"Median of A: {median_A}")    print(f"Standard
Deviation of A: {std_dev_A}")
               print("Bitwise AND:", ' '.join(map(str, and_result)))
print("Bitwise OR:", ' '.join(map(str, or_result)))         print("Bitwise
XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

Output:



3.2.5 Numpy: Copying and Viewing Arrays

Problem Statement:

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view a ects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

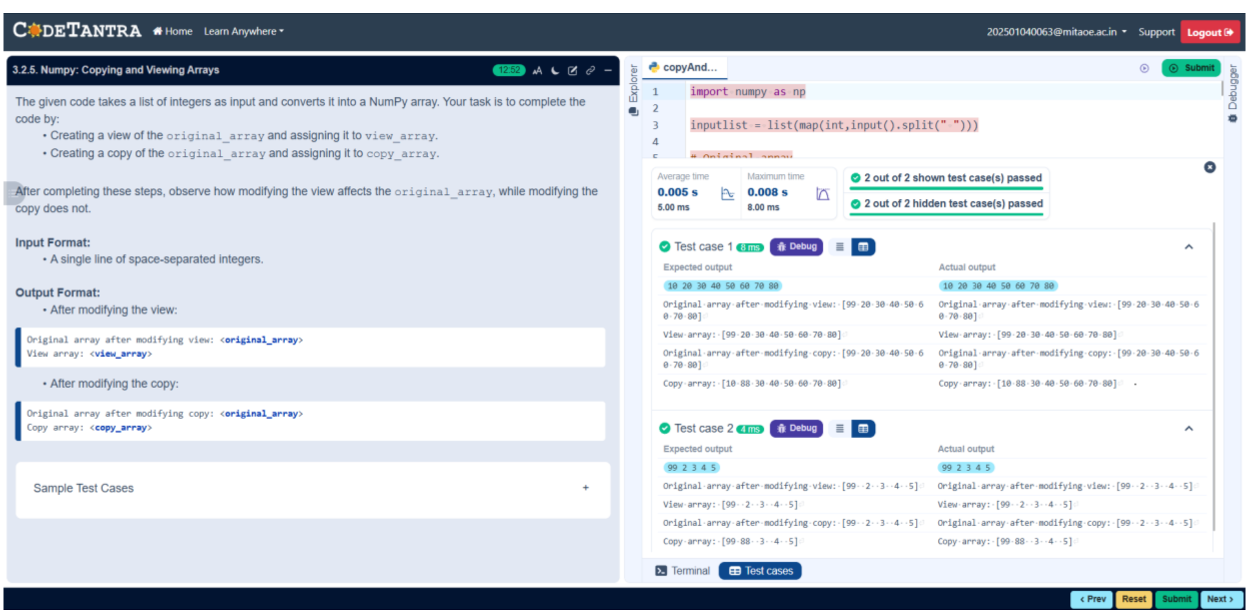
Output Format:

- After modifying the view:
  Original array after modifying view: <original_array>
  View array: <view_array>
- After modifying the copy:
  Original array after modifying copy: <original_array> Copy array: <copy_array>

Code:

```
import numpy as np
inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy copy_array[1]
= 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

Output:



3.2.6 Searching, Sorting, Counting, Broadcasting

Problem Statement:

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where search_value appears in array1 and print these indices.
2. Counting: Count how many times count_value appears in array1 and print the count. 3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.

4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.

2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

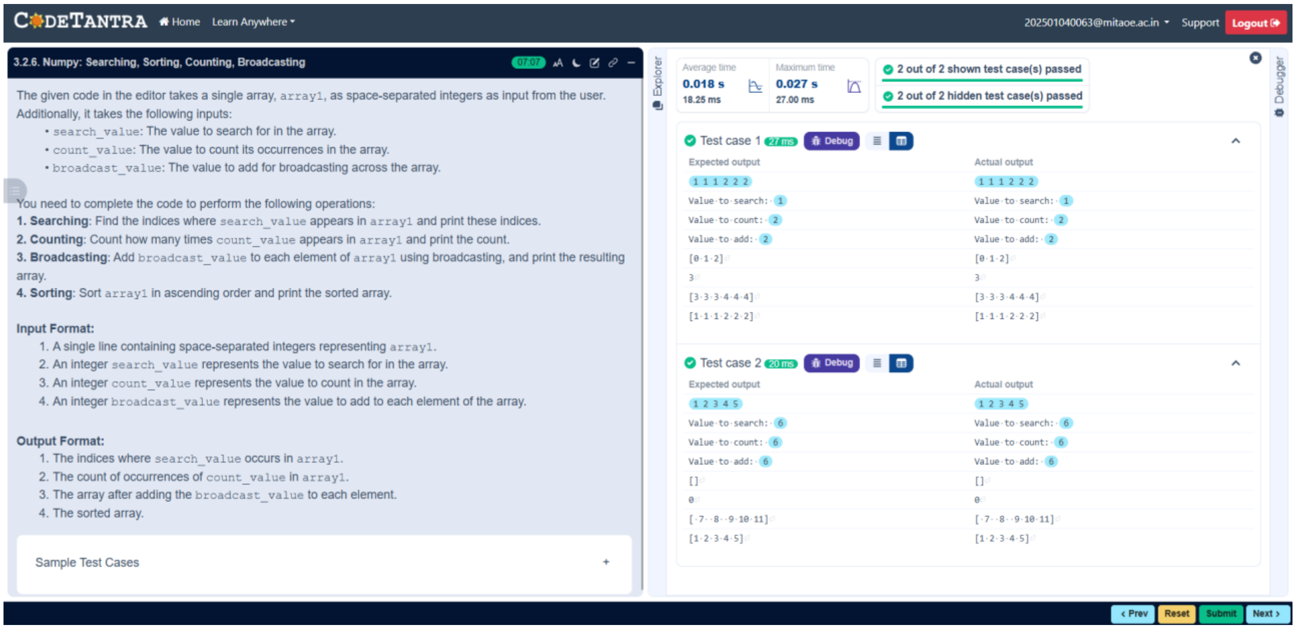
Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: ")) count_value
= int(input("Value to count: ")) broadcast_value =
int(input("Value to add: "))

# Find indices where value matches in array1 a
= np.where(array1 == search_value)[0] print(a)

# Count occurrences in array1 b =
np.count_nonzero(array1 == count_value)
print(b)

# Broadcasting addition c =
array1 + broadcast_value
print(c)

# Sort the first array d
= np.sort(array1)
print(d)
```

Output:



3.2.7 Student Data Analysis and Operations

Problem Statement:

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.

- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.
- Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.
- Find the count of students who scored >=90 in each subject: Count the number of students who scored 90 or more marks in each subject.
- Find the count of subjects in which each student scored >=90: Determine how many subjects each student scored 90 or more marks in.
- Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.
- Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.
- Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Code:

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

roll = a[:, 0].astype(float)
s1 = a[:, 1] s2 = a[:, 2] s3
= a[:, 3]

# 1. Print all student details print("All
student Details:\n", a)

# 2. print total students print("Total
Students:", len(a))
```

```
# 3. Print all student Roll numbers
print("All Student Roll Nos", roll)

# 4. Print subject 1 marks
print("Subject 1 Marks", s1)

# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(s2))

# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(s3))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. print Total marks of students
total = s1 + s2 + s3 print("Total
Marks", total)

# 9. print average marks of each student
print(np.round(total/3, 1))

# 10. print average marks of each subject
print("Average Marks of each subject", np.mean(a[:, 1:], axis = 0))

# 11. print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean([s1, s2], axis = 1))

# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean([s1, s3], axis = 1))

# 13. print Roll number who got maximum marks in Subject 3
print("Roll no who got maximum marks in Subject 3", float(roll[np.argmax(s3)]))

# 14. print Roll number who got minimum marks in Subject 2
print("Roll no who got minimum marks in Subject 2", float(roll[np.argmin(s2)]))

# 15. print Roll number who got 24 marks in Subject 2
print("Roll no who got 24 marks in Subject 2",roll[s2 == 24].astype(float).reshape(1, -1))

# 16. print count of students who got marks in Subject 1 < 40
print("Count of students who got marks in Subject 1 < 40", np.sum(s1 < 40))

# 17. print count of students who got marks in Subject 2 > 90
```

print("Count of students who got marks in Subject 2 > 90:", np.sum(s2 > 90))

# 18. print count of students in each subject who got marks >= 90
print("Count of students in each subject who got marks >= 90:", np.sum(a[:, 1:] >= 90, axis = 0))

# 19. print count of subjects in which each student got marks >= 90 print("Roll no:", roll)
print("Count of subjects in which student got marks >= 90:", np.sum(a[:, 1:] >= 90, axis = 1))

20. Print S1 marks in ascending order print(np.sort(s1))

# 21. Print S1 marks >= 50 and <= 90 print(a[(s1 >= 50) & (s1 <= 90)].astype(float))
print(a.astype(float))

22. Print the index position of marks 79 print(np.where(s1 == 79))

Output: